

Smart Anti-Theft Security System



ANTI-THEFT SECURITY

Security on working



System Overview

This project is a smart vehicle anti-theft system designed to protect cars or motorcycles from unauthorized access, while allowing the vehicle owner to monitor the car remotely from home using a web-based dashboard

The system:

- Detects suspicious activity around the vehicle
- Triggers alarms when theft is detected
- Prevents unauthorized access by locking the vehicle doors
- Sends alerts and allows monitoring and control through a dashboard
- 🎯 Main Objective:

Detect → Analyze → React → Notify

Inputs and Outputs Explained

Ultrasonic Sensor

The ultrasonic sensor detects motion by emitting sound waves and measuring the time taken for the waves to return, ensuring immediate awareness of unauthorized access near the vehicle.

Buzzer

Activates alarm

Microphone Sensor

A microphone sensor listens for a specific **two-clap code**, allowing the system to recognize authorized users. This feature enhances security by preventing unauthorized access attempts.

LED

Shows system state

Servo Motor

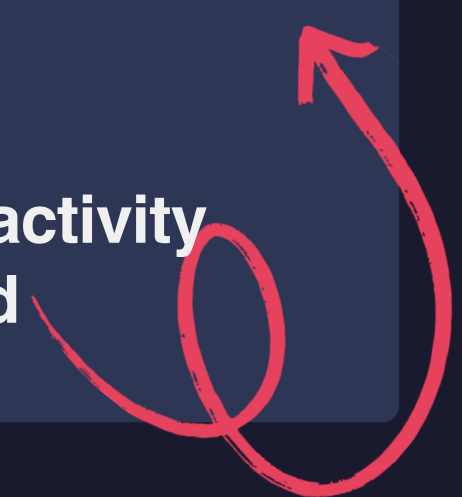
The servo motor acts as a locking mechanism, responding to the sensor inputs. It unlocks when the correct code is detected, ensuring security while providing easy access for authorized users.

LED Indications:

☐ Green → Safe mode

☐ Yellow → Suspicious activity

● Red → Theft detected



How It Works

Detect Motion

The system begins by using an **ultrasonic sensor** to detect any motion near the vehicle. This initiates the security protocol and prepares for potential activation.

Listen for Claps

Next, the microphone sensor actively listens for a **clapping code**. This unique sequence allows the system to recognize an authorized user and unlock the vehicle.

Correct Code Unlocks

Upon detecting the correct clapping code, the system activates the **servo motor**, unlocking the vehicle's doors and providing access to the authorized user.

Incorrect Triggers Alarm

If the system detects an incorrect sequence or no claps, it activates the **LED alert** and the buzzer, keeping the vehicle secure and deterring potential theft.

System Scenarios

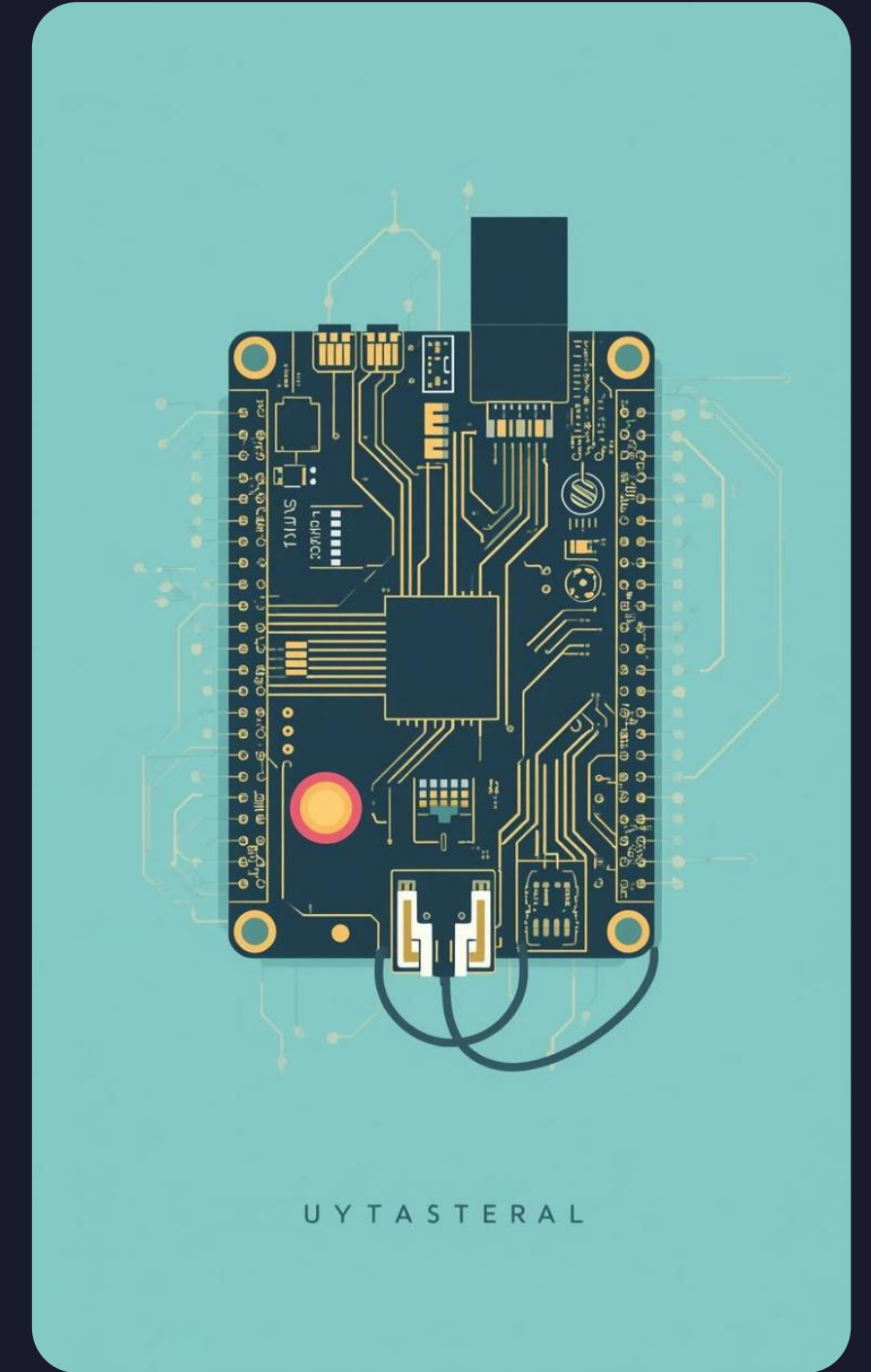
Scenario 1: Suspicious Activity

- Motion detected
 - Person is close but no tampering
- Warning mode
- Dashboard shows distance, time and turn on Yellow LED

- cenario 2: Theft Attempt
 - Motion detected
 - Very close distance for a long time
 - Loud sound (tampering)
- Alarm activated
- Vehicle doors locked
- Notification sent to the dashboard

Advantages of the Anti-Theft System

- Low-cost solution
- Real-time monitoring from home
- Preventive security approach
- Easy installation
- Scalable and upgradable

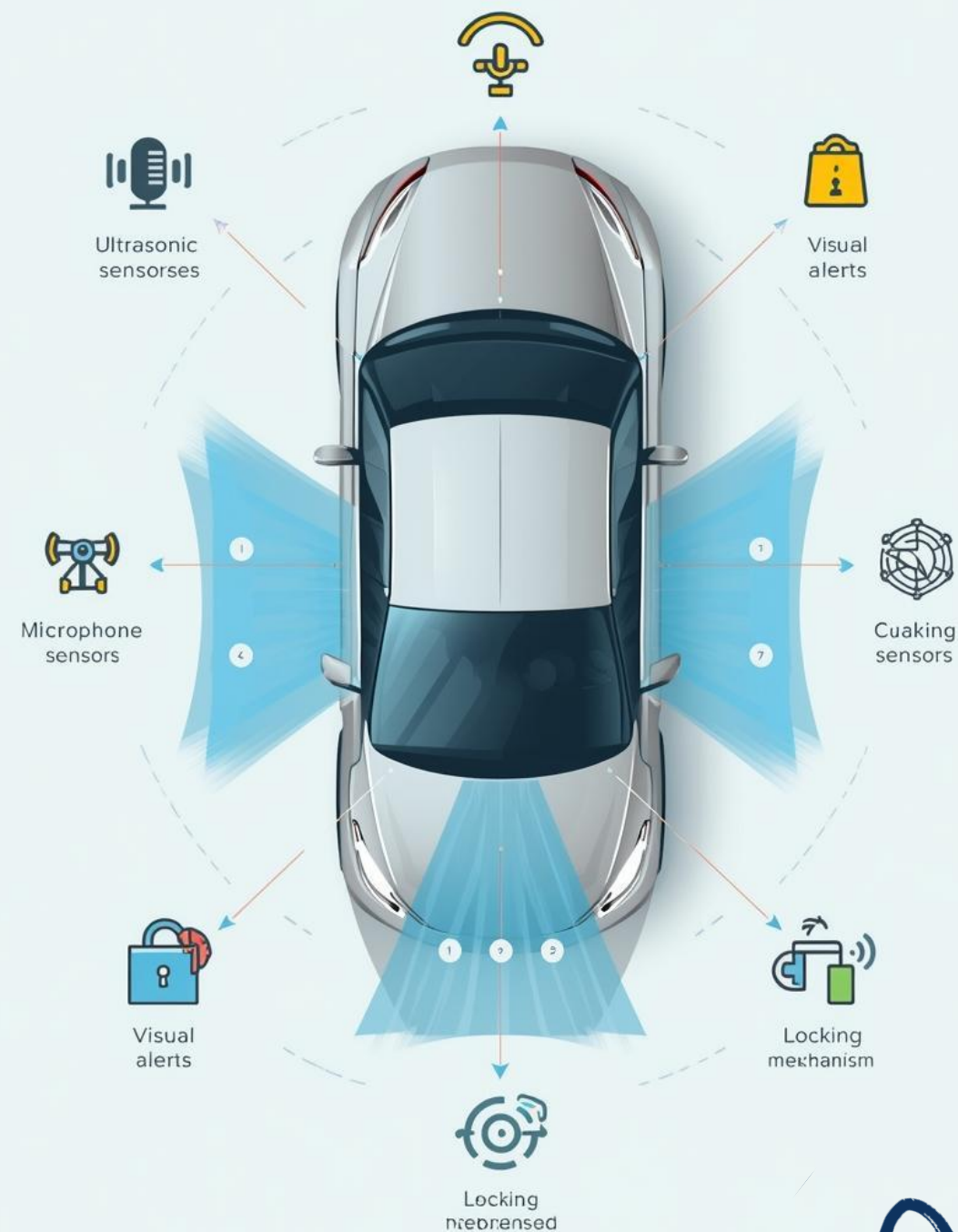


Future Enhancements

- **GPS tracking**
- **Camera integration**
- **Machine learning-based behavior analysis**
- **RFID or fingerprint authentication**

Car Safety Sensor

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Thank You

We appreciate your attention